



AEROSPACE CONTROL SYSTEMS

PRACTICAL SESSION 1

TRANSFER FUNCTION AND STATE SPACE MODELING AND ANALYSIS

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Exercise 1

Given the system:

$$G(s) = \frac{1}{s^2 + 5s + 6} \begin{bmatrix} 1 & 1 \\ s + 1 \end{bmatrix}$$

Obtain the state space representation.

Solution:

Using the `tf2ss` function in MATLAB, the following state space matrices are obtained. As indicated by the dimensions (A is 2×2 , B is 2×1 , C is 2×2 and D is 2×1), this system has 2 states, 1 input $u(t)$, and 2 outputs.

$$A = \begin{bmatrix} -5 & -6 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$C = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Exercise 2

From the state space representation obtained previously, obtain the poles and zeros of the transfer function of the system.

Solution:

Evaluating the matrices with `ss2zp`, the resulting poles and zeros are:

$$Poles : p = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$$
$$Zeros : z = \begin{bmatrix} \infty \\ -1 \end{bmatrix}$$

Note: The zero at infinity returned by MATLAB means that its transfer function approaches zero as $s \rightarrow \infty$. The system only has one finite zero at $s = -1$.

Exercise 3

Given the system defined by:

$$A = \begin{pmatrix} -5 & -6 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$
$$C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

Obtain the transfer function of the system.

Solution:

By performing a dimensional analysis on the matrices, we can observe that since B is a 2×2 matrix, the system has 2 inputs. Since C is a 1×2 matrix, it only has 1 output. Because of these 2 inputs, there are two transfer functions, which are evaluated with `ss2tf` (one for each input).

$$TF_1(s) = \frac{-6}{s^2 + 5s + 6}$$
$$TF_2(s) = \frac{s - 6}{s^2 + 5s + 6}$$

Exercise 4

Obtain the diagonal and observable canonical forms of the system $G(s) = \frac{1}{s^2+5s+6} \begin{bmatrix} 1 \\ s+1 \end{bmatrix}$, obtaining the transformation matrix T in each case.

Solution:

The transformation matrix T returned by MATLAB's `canon` function corresponds to the inverse of the theoretical transformation matrix, meaning the transformed state matrix is computed as $\tilde{A} = T_{matlab} \cdot A \cdot T_{matlab}^{-1}$.

Diagonal (Jordan) form:

$$\begin{aligned} A_{diag} &= \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}, & B_{diag} &= \begin{bmatrix} -3.6056 \\ -2.8284 \end{bmatrix} \\ C_{diag} &= \begin{bmatrix} 0.2774 & -0.3536 \\ -0.5547 & 0.3536 \end{bmatrix}, & D_{diag} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ T_{diag} &= \begin{bmatrix} -0.8321 & 0.7071 \\ 0.2774 & -0.3536 \end{bmatrix} \end{aligned}$$

Observable form:

$$\begin{aligned} A_{obs} &= \begin{bmatrix} 0 & -6 \\ 1 & -5 \end{bmatrix}, & B_{obs} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ C_{obs} &= \begin{bmatrix} 0 & 1 \\ 1 & -4 \end{bmatrix}, & D_{obs} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ T_{obs} &= \begin{bmatrix} 1 & -5 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Exercise 5

Given the system

$$G(s) = \frac{1}{s^4 + 12s^3 + 55s^2 + 124s + 120}$$

obtain the poles and sort them. Represent the poles and zeros graphically.

Solution:

Evaluating the transfer function yields an empty 0×1 array for the zeros, indicating that the system has no zeros. The `esort` function is used to automatically organize the obtained poles by their real parts.

The sorted poles of the system are:

$$p_1 = -2 + 2i, \quad p_2 = -2 - 2i, \quad p_3 = -3, \quad p_4 = -5.$$

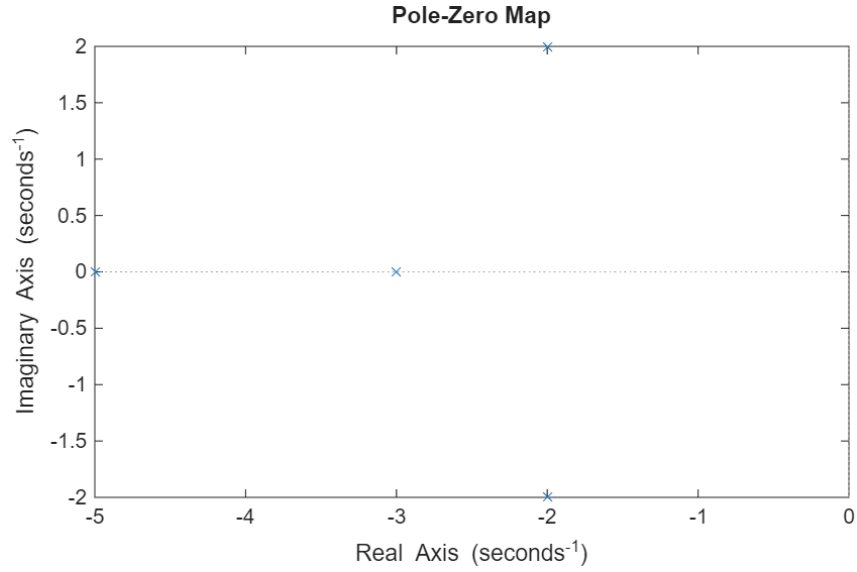


Figure 1: Pole-Zero map of the system.

Exercise 6

Given the system $G(s) = \frac{1}{s^2+5s+6} \begin{bmatrix} 1 \\ s+1 \end{bmatrix}$ obtain the state space representation and represent the impulse and step responses.

Solution:

Using the matrices defined in Exercise 1, the step and impulse responses are computed using **step** and **impz** functions:

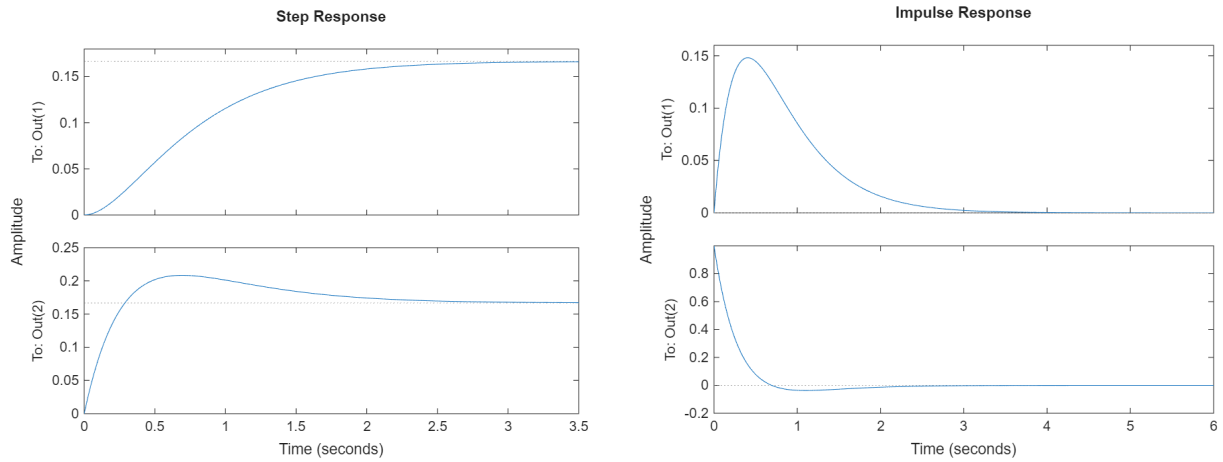


Figure 2: Step (left) and Impulse (right) responses.

Exercise 7

Obtain the response of the previous system to a ramp.

Solution:

Using the `lsim` function for an input vector $u = t$:

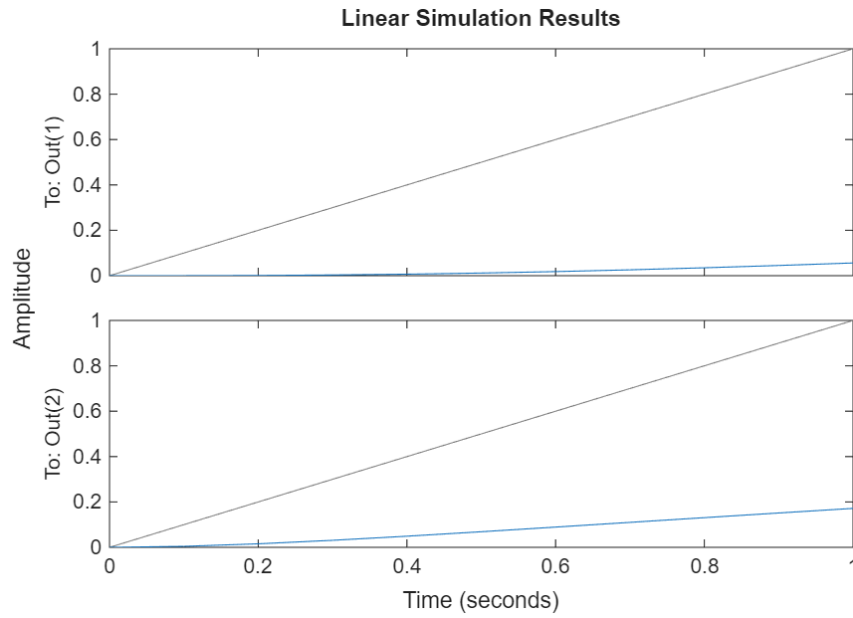


Figure 3: System response to a ramp input.

Exercise 8

Obtain the Bode and Nyquist plots of the previous system.

Solution:

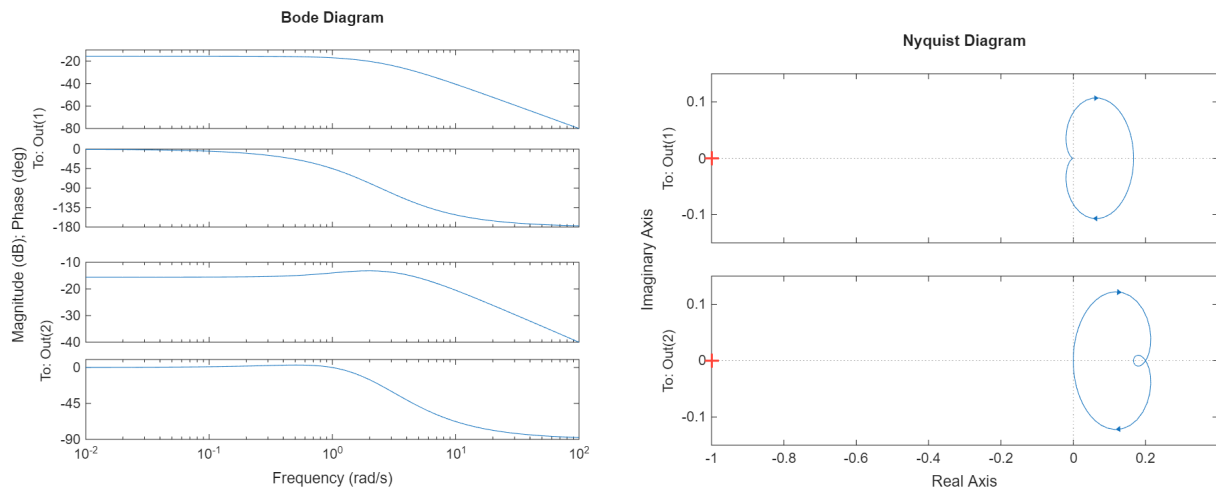


Figure 4: Bode plots (left) and Nyquist plot (right).